

Question	Answer	Marks
5(a)(i)	west to east	B1
5(a)(ii)	above the Equator	B1
5(a)(iii)	value in range $(1-300) \times 10^9$ Hz	A1
5(b)(i)	gain/dB = $10 \lg(P_2/P_1)$	C1
	$-195 = 10 \lg(P/3000)$ or $195 = 10 \lg(3000/P)$	C1
	power = 9.5×10^{-17} W	A1
5(b)(ii)	up-link has been (greatly) attenuated (before reaching satellite) or down-link signal must be (greatly) amplified (before transmission back to Earth) or up-link has (much) smaller intensity/power than down-link	B1
	(different frequency) prevents down-link (signal) swamping up-link (signal)	B1

Question	Answer	Marks
6(a)	frequency of light	B1